

Teach yourself touch-typing in two hours

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February 24, 2009

What is this text for?

“Touch-typing” means typing on a computer keyboard (or a typewriter) *without looking* at the keyboard.

I would like to help you teach yourself touch-typing using an innovative method. The method is very fast and literally requires only two hours of exercise. You *do not* need an actual keyboard — you can learn and practice touch-typing using just paper. After going through the seven lessons (this takes about half an hour to one hour), you will be able to touch-type with the speed of about 60 keystrokes per minute.¹ Typing only 60 keystrokes per minute (1 keystrokes per second) is, of course, very slow, and you will need more practice. After another hour or so of practice, you will increase the speed of your typing to about 200 keystrokes per minute. If you have played the piano or another keyboard instrument, your final typing speed will be faster since your fingers are already used to performing finely controlled movements.

Here I show exercises geared to the standard American QWERTY keyboard. However, the same method can be adapted to any other keyboard layout. The author has learned two keyboard layouts (the Russian typewriter and the American QWERTY) and retained the use of both layouts after many years. If you want to learn another layout, you will have to prepare your own exercises, which is not difficult if you can edit a little L^AT_EX. The L^AT_EX sources this document are attached to this PDF file.

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Lesson 1: General ideas. Left hand

For now, we ignore the punctuation keys, special characters, and digits; we concentrate on the letters. The keyboard has a strange slanted shape; the letters are not quite on top of each other. However, you should imagine that the keyboard is a *rectangular* array of letters, 3×10 . Each vertical row has three letters, except for some rows at far right. The rows read **QAZ**, **WSX**, **EDC**, **RFV**, **TGB**, **YHN**, **UJM**, **IK**, **OL**, **P**. (Remember, the punctuation keys are ignored for now.)

The idea of touch-typing can be summarized as follows:

- You type the letter keys with *eight* fingers (thumbs are used only to press the **space bar**).

¹Typing speed is sometimes measured in “words per minute” but this is ambiguous, so I talk about keystrokes per minute, that is, the number of all keystrokes including letters, digits, spaces, and punctuation.

- Each of the eight fingers is responsible for typing letters belonging to some *vertical row*.

To avoid any misunderstanding, we will be counting the fingers of each hand from 1 to 5 by starting from the thumb, which will be the finger number 1.

From this point and until Lesson 7 you *do not need* a keyboard.

Left hand

The finger number 5 of the left hand (also called the little finger, or the “pinky”) rests in the vertical row **QAZ**. The finger number 4 (also called the “ring finger”) of the left hand rests in the vertical row **WSX**. The finger number 3 (the “middle finger”) of the left hand rests in the vertical row **EDC**. The finger number 2 (the “index finger”) of the left hand rests in the vertical row **RFV**.

You now need to memorize the four nonsensical words: **qaz**, **wsx**, **edc**, and **rfv**. They correspond to the four fingers of your left hand, counted left to right starting from the little finger. Use your imagination to memorize these words. Invent some funny pronunciation for them. For example, pronounce these words to yourself like this:

qaz	<i>quasi</i>
wsx	<i>wessex</i>
edc	<i>educ</i> (short for “education”)
rfv	<i>are-fever</i>

To make sure you memorize these four words and associate them with fingers of your left hand, repeat these words to yourself in the order of your fingers, each time tapping that finger. Perform two exercises: First, tap with the finger that corresponds to a given word. Second, tap with some finger and say aloud the corresponding word.

Lesson 2: Right hand

The finger number 2 (the “index finger”) of the right hand rests in the vertical row **UJM**. The finger number 3 (the “middle finger”) of the right hand rests in the vertical row **IK**. The finger number 4 (the “ring finger”) of the right hand rests in the vertical row **OL**. The finger number 5 (the “little finger”) of the right hand rests in the vertical row **P**. (Well, this is a very short row, but the finger number 5 will eventually have more work to do.)

You now need to memorize the four nonsensical words: **ujm**, **ik**, **ol**, and **p**. They correspond to the four fingers of your right hand, counted left to right starting from the index finger. Use your imagination to memorize these words. Invent some funny pronunciation for them. For example, pronounce these words to yourself like this:

ujm	<i>you-jump</i>
ik	<i>eye-ck</i>
ol	<i>ole</i> (short for “old”)
p	<i>pee</i>

To make sure you memorize these four words and associate them with fingers of your right hand, repeat these words to yourself in the order of your fingers, each time tapping that finger. Perform two exercises: First, tap with the finger that corresponds to a given word. Second, tap with some finger and say aloud the corresponding word.

Lesson 3: Index fingers moving sideways

Each of the the index fingers actually services *two* rows. The index finger of the left hand normally rests within the **RFV** row but will move a little to the right in order to press any of the keys in the **TGB** row. The index finger of the right hand will normally be within the **UJM** row but will move a little to the left in order to press any of the keys in the **YHN** row.

You now need to memorize the words **tgb** and **yhn**. Pronounce them, for example, like this:

tgb	<i>tee-gee-bee</i> (not “KGB” and not “TGV” but close)
yhn	<i>yeah-hon</i> (short for “yes, honey”)

Now you can practice all the possible 10 words. Repeat these words to yourself, each time tapping that finger, in the order of your fingers. Note that you need to move the left index finger to the right when you tap “**tgb**” (and the right index finger to the left when you tap “**yhn**”) and then *you need to return the index finger* to its usual place. Do not let an index finger stay in a sideways shifted position!

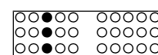
Perform two exercises: First, tap with the finger that corresponds to a given word. Second, tap with some finger and say aloud the corresponding word.

Lesson 4: Practicing the vertical rows

You have by now memorized the words **qaz**, **wsx**, **edc**, **rfv**, **tgb**, **yhn**, **ujm**, **ik**, **ol**, and **p**. Now you will practice all the ten words in random order. This is most easily done with special exercise sheets (see Exercise Sheets 1 and 2).

Sheet 1

Sheet 1 has many rectangles; each rectangle contains a schematic diagram of the ten vertical rows. Each row corresponds to one finger of one hand. (As explained before, the index fingers are responsible for *two* rows each.) In every little diagram, one of the vertical rows is filled in black, indicating one of the fingers. Place your fingers on a table top. For each diagram, you need to tap the finger indicated and then you need to say aloud the corresponding word. For instance, the diagram



means the left finger 3, so you tap that finger and say **edc**. The diagram



means the right index finger shifted to the left, so you move the right index finger a little to the left, tap, say **yhn**, and return the finger to its normal position.

If you had to move the index finger to the side when you tapped, *you need to return the index finger to its normal position right after you tapped it*. Do not let an index finger stay in a sideways shifted position!

If you cannot recall the word corresponding to some finger, maybe you can remember the word for a nearby finger? This may help recalling the right word.

Sheet 2

Sheet 2 has many rectangles; each rectangle contains a word (drawn vertically). You need to tap the finger corresponding to that word. For example, if you see



then you need to tap the left finger 3 (the middle finger). Again, it is important that you return the index fingers back to their normal positions after moving them sideways. For example, when you see



you need to extend the left index finger a little to the right, tap, and then immediately return the index finger to its normal position.

There are four versions of Sheet 1 and four versions of Sheet 2; each time the diagrams were generated randomly. You should practice first Sheet 1, then Sheet 2, then another Sheet 1, then another Sheet 2, and so on. You can also practice a half of Sheet 1, then a half of Sheet 2, then the second half of Sheet 1, then the second half of Sheet 2.

Going through one complete sheet takes about 5 minutes.

Practice no more than four sheets a day.

The mirror exercises

If you have more time to practice, here is another set of exercises that can be performed with the same sheets 1 and 2.

For each diagram in Sheet 1, tap the finger that is *in the mirror symmetric position* and say aloud the corresponding word. For example, left finger 3 is mirror symmetric to right finger 3. So if you see the right finger 3 marked, you tap the left finger 3 and say **edc**.

For each word in Sheet 2, tap the finger that is *in the mirror symmetric position* to that normally corresponding to that word. For example, **edc** normally corresponds to the left finger 3, so if you see **edc** then you tap with the *right* finger 3.

Lesson 5: Looking at letters and typing rows

This exercise is performed with Sheet 3.

Sheet 3 contains random letters. You need to look at each letter and say aloud the word in which this letter occurs. For example, if you see the letter **G** then you say **tg**b**** because this is the word that contains **G**. Then you need to tap the finger responsible for the word **tg**b**** (this is the left index finger shifted to the right). Repeat the same for each letter in Sheet 3.

There are four versions of Sheet 3, each randomly generated. Do not practice Sheet 3 many times in a row; instead, practice a half of Sheets 1 and 2 and then Sheet 3. Although practice sessions are very short, you could leave several hours between practicing sessions.

Lesson 6: Actually typing letters

This exercise is again performed with Sheet 3. This time, you need to move each finger a little more precisely.

Sheet 3 contains random letters. You need to look at each letter, determine the word in which this letter occurs, determine the finger responsible for that word, and tap the finger. However, during this lesson you will need to move each finger a little more precisely. If the letter is at the beginning of the word, you need to shift your finger a little upwards when you tap. If the letter is the third letter of the word, you need to shift your finger a little down when you tap. If the letter is the second letter of the word, you tap the finger in its normal position. If you are tapping with an index finger in a sideways position, you need to shift additionally to the sideways shifting.

For example, if you see **W** you determine that it belongs to **wsx**, and you determine that **wsx** is tapped with the left finger 4. Since **W** is the first letter in **wsx**, you shift the left finger 4 a little upwards and then tap it. *Then you move the left finger 4 back to its normal position.* After that you go on to the next letter.

In this exercise you will be shifting fingers up and down as you tap. It is very important to return every finger to its normal position immediately after tapping. Never leave any finger in a shifted position!

You need to go through the four versions of Sheet 3. Do not practice Sheet 3 many times in a row; instead, practice a half of Sheets 1 and 2 each, and then Sheet 3. Although practice sessions are very short, you could leave several hours between practicing sessions. Turn Sheet 1 upside-down to get a different exercise sheet.

Lesson 7: Take back the keyboard

Now you are ready to type on a real keyboard.

The position of the hands

The index fingers (that is, the fingers number 2) are positioned on the keys **F** and **J**; note that these keys have small indentations — these are made to help you put the index fingers in the correct position by touching the keyboard, that is, *without looking at the keyboard*. After you find the two indentations, put your middle fingers next to the index fingers in the horizontal row, and put all other fingers onto the neighboring keys in the middle horizontal row. This is the normal position of your fingers while typing. The wrists must be high over the keyboard, and the eight “working” fingers must lightly touch the eight keys **A, S, D, F, H, J, K, L**. It is quite important to maintain this position of all the fingers while typing.

The Shift keys

The two **Shift** keys are pressed by extending the left and right fingers 5 (the little fingers). If you need to press **Shift-A**, you deduce that **A** belongs to **qaz**, which is on the finger 5 in the left hand, and so it is convenient to press the **Shift** key with the finger 5 of the *right* hand together with **A**.

Diving in

Take some short text and put it in front of you and/or to the left of the keyboard. You do not need to look at the keyboard! Look into the text instead. Look at the first letter of the text. For example, if the text is “**A quick brown fox jumped over the lazy dog**” the first letter could be the capital “**A**.” Determine the word that contains this letter; this is **qaz** and is typed with the left finger 5. The **Shift** key is therefore pressed with the right finger 5. Since **A** is the *second* letter in **qaz**, the left finger 5 remains in its normal position and presses the key. Done! Move the right finger 5 back to its normal position. Now go on to the second letter. Continue until your text is typed.

The **Space bar** is pressed with either of the thumbs.

The more you practice, the faster your typing will become. Remember:

- Never look back at the keyboard!
- Return every finger to its normal position!

Digits, punctuation, other keys

Digits are typed with the same eight fingers extended all the way up. It is not difficult to figure out how to type digits: the left fingers 5 to 2 type the digits 0 to 4, and the right fingers 2 to 5 type the digits 5 to 9.

Punctuation keys (`.`, `,`, `;`, `-`, `?`, `!`, `'` and so on), as well as any special keys such as `@` or `#`, need to be memorized separately. Many often used keys, like the period (right finger 3, shifted down) and the comma (right finger 4, shifted down), will be memorized quickly once you start typing. If you need to type a lot of special symbols, you will find that you automatically remember their positions with practice.

Final comments

The method presented here is radically different from the traditional method. The traditional method is to make you type **a-f-a-f** and so on, until you slowly learn the keyboard, one letter key at a time. In the present method, you first need to learn those funny short words — **qaz**, **wsx**, and so on — and then you just need to use your imagination to control your fingers. It is perfectly possible to learn and to practice this method without an actual keyboard.

The history of my acquaintance with the method is the following. There was a researcher in psychology whose name I never knew; her thesis was that it is much easier for humans to learn and remember *words* — even short and nonsensical words like “**wsx**” — than to remember *individual letters*, and furthermore that the human brain can manipulate such short words much easier than individual letters. To test this hypothesis, she designed a course of touch-typing for school children, very much like the one I present here. One of her friends organized this class to be given in a local high school on a volunteer basis. I was one of the subjects in her class. Everyone in the class learned to touch-type in four half-hour lessons. In that class we learned the keyboard layout of a Russian typewriter; later I used the same method to teach myself the QWERTY layout. Indeed, it appears to be much easier and quicker to learn short words than to learn one letter at a time. During the practice exercises, the memorized short words *reinforce* the learning of the keyboard layout in a way that is not available with the traditional method.

Obtaining the L^AT_EX source of this document

I produced this text with the L^AT_EX and L^AT_EX document processing systems. You may distribute this text according to the GNU Free Documentation License. (Look it up on the Web if you need to see the precise terms of that license.) In particular, you are entitled to modify this document. If you distribute a modified version of this text, you must distribute *under the terms of the same license*. In other words, you

cannot prohibit other people from distributing your modified versions or from modifying them further.

In order to modify the document, you need the L^AT_EX sources. The sources are bundled in the PDF file as a **tar.gz** archive. To extract the sources, you may use the free utility **pdftk** and type the following command:

```
pdftk touch_typing.pdf unpack_files
```

where **touch_typing.pdf** is the name of this PDF file. The archive **touch_typing.tar.gz** will be created in the current directory.

If you want to change the source and to rebuild this PDF file, you need to execute the following commands (long lines were broken by the backslash `\` symbol):

```
pdflatex typing.tex
```

```
pdflatex sheets.tex
```

```
tar zcvf touch_typing.tar.gz typing.lyx \
typing.tex sheets.tex sheets.pdf random.tex \
arrayjob.sty multido.sty multido.tex
```

```
pdftk typing.pdf sheets.pdf cat \
output touch_typing.pdf
```

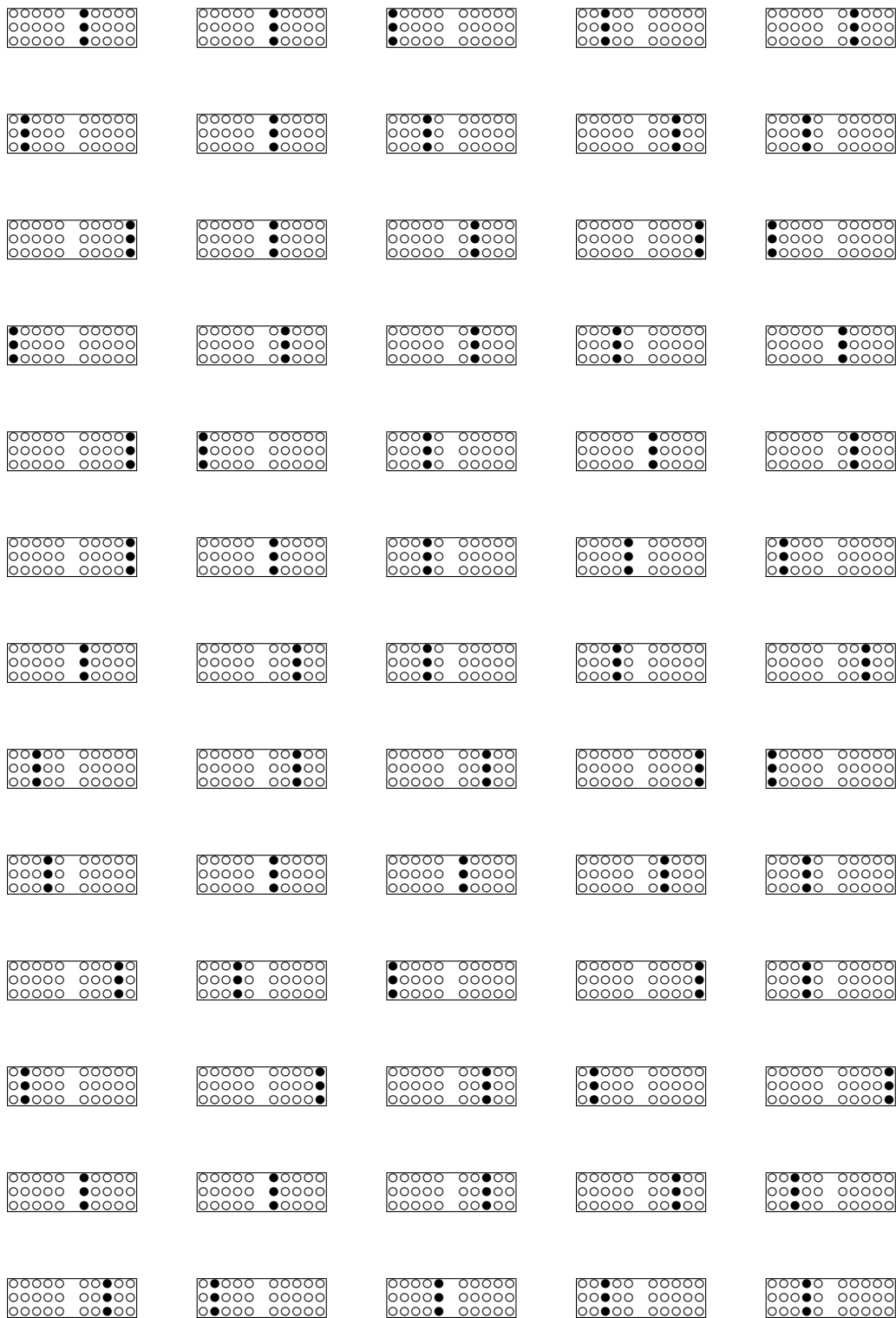
```
pdftk touch_typing.pdf attach_files \
touch_typing.tar.gz output touch_typing_guide.pdf
```

These commands are also given in the **Makefile**, so you can execute them by running **make**.

In order to adapt the exercises to the keyboard layout, you need to edit the file **sheets.tex** and modify the arrays named **Rows** and **Letters** at the beginning of that file. Examples are given there for the German keyboard layout.

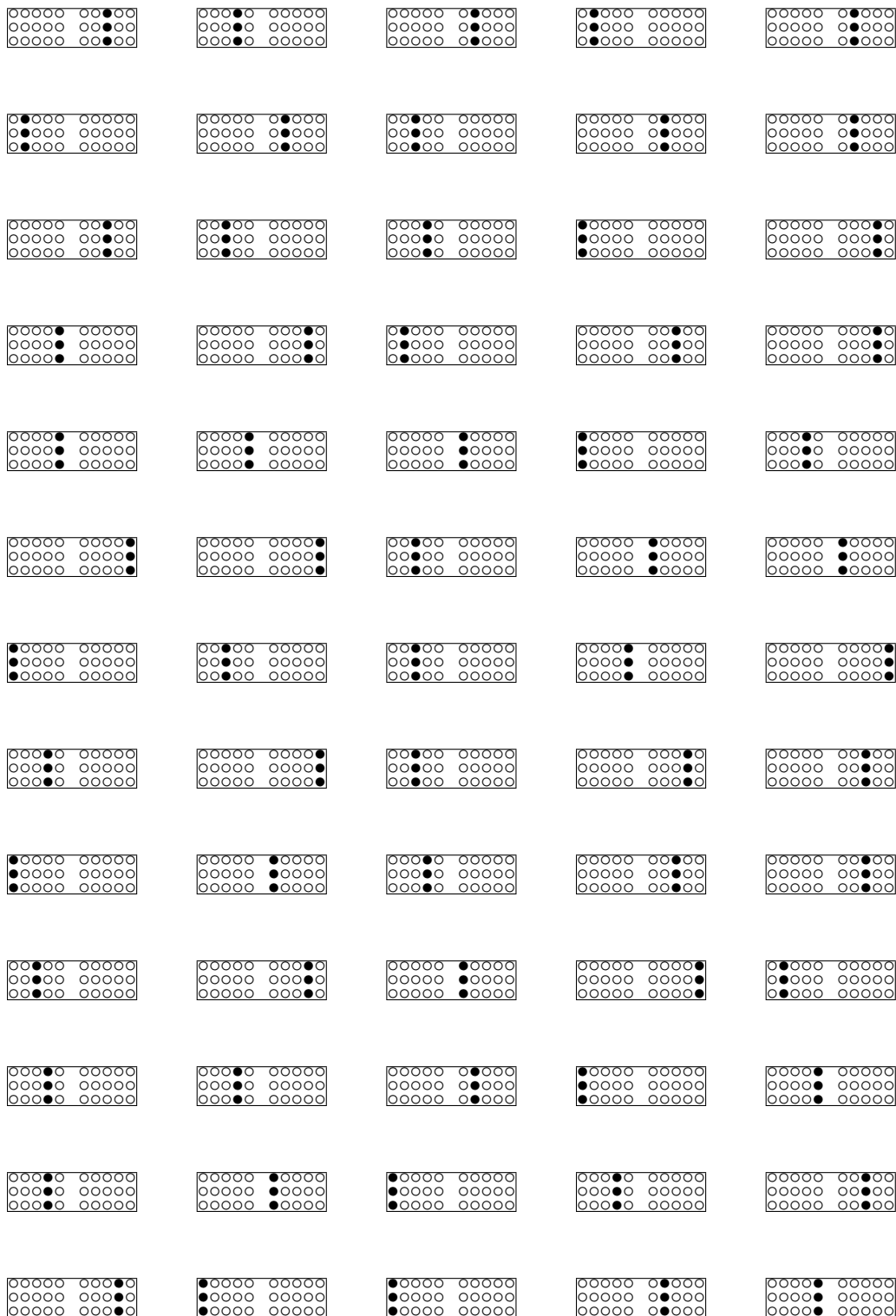
Sheet 1

For each of the shown fingers, you should pronounce the corresponding word to yourself.



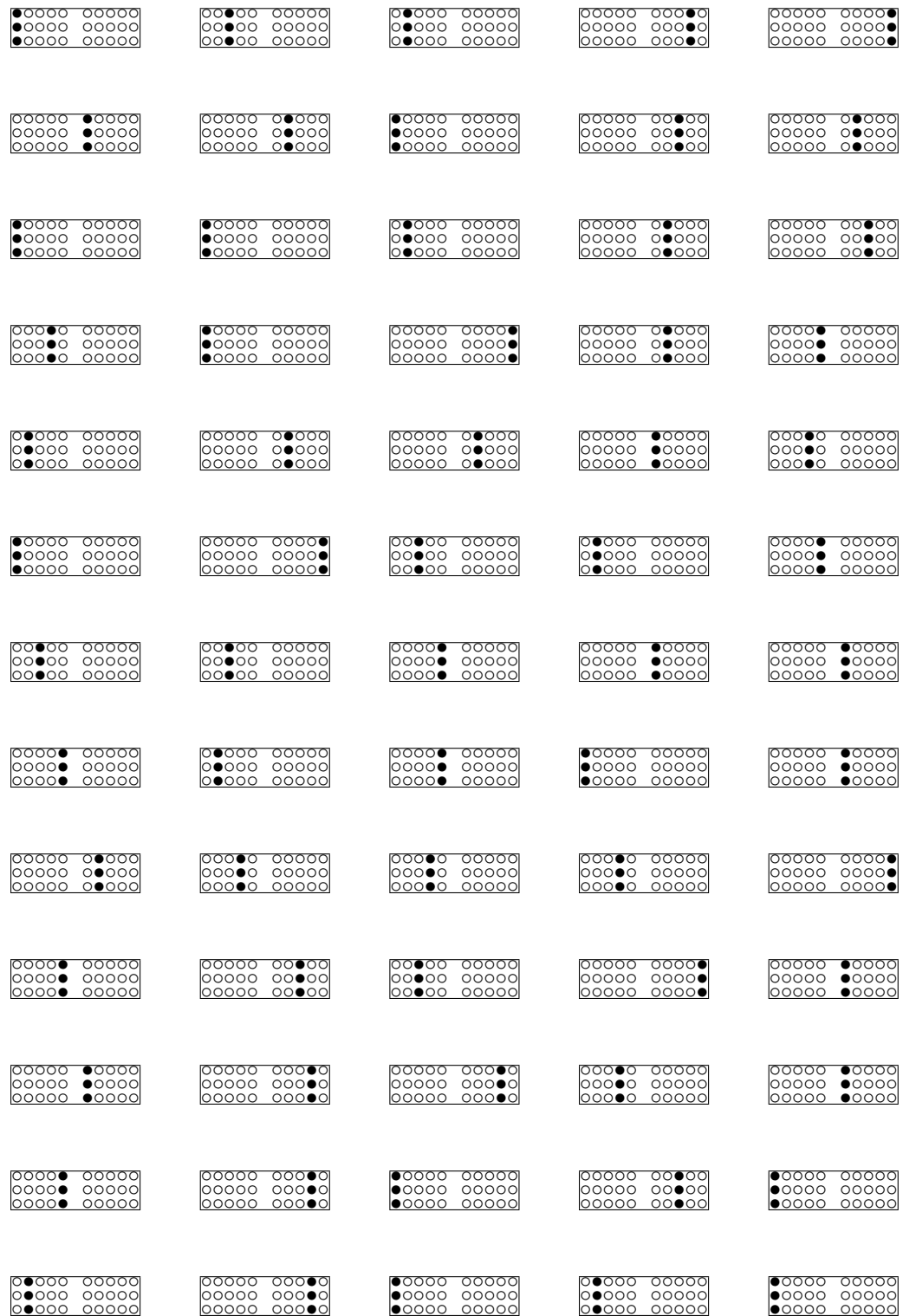
Sheet 1

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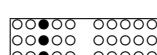
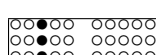
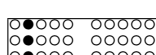
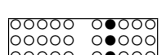
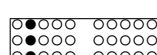
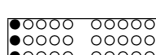
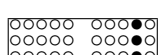
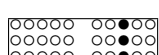
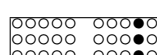
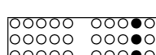
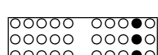
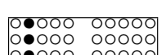
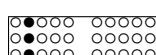
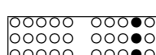
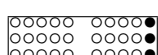
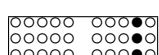
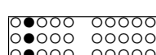
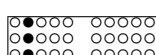
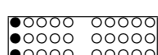
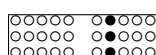
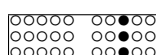
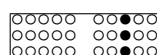
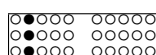
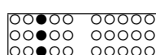
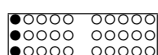
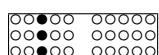
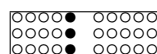
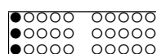
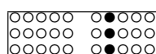
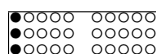
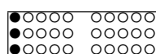
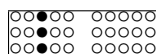
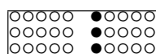
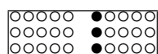
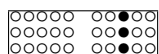
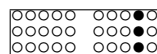
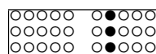
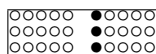
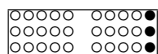
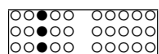
Sheet 1

For each of the shown fingers, you should pronounce the corresponding word to yourself.



Sheet 1

For each of the shown fingers, you should pronounce the corresponding word to yourself.



Sheet 2

For each of the given words, you should lift and tap the corresponding finger.

E
D
C

Q
A
Z

T
G
B

P

W
S
X

Q
A
Z

R
F
V

R
F
V

I
K

Q
A
Z

O
L

Y
H
N

W
S
X

Q
A
Z

U
J
M

Q
A
Z

P

Y
H
N

U
J
M

U
J
M

O
L

E
D
C

T
G
B

U
J
M

R
F
V

O
L

W
S
X

O
L

T
G
B

E
D
C

Y
H
N

W
S
X

T
G
B

T
G
B

W
S
X

P

Q
A
Z

O
L

P

R
F
V

P

I
K

Q
A
Z

Q
A
Z

U
J
M

R
F
V

O
L

T
G
B

I
K

W
S
X

T
G
B

Y
H
N

T
G
B

R
F
V

T
G
B

W
S
X

R
F
V

U
J
M

W
S
X

O
L

T
G
B

W
S
X

W
S
X

I
K

R
F
V

R
F
V

E
D
C

Y
H
N

P

R
F
V

Y
H
N

I
K

Y
H
N

Y
H
N

W
S
X

U
J
M

Q
A
Z

E
D
C

R
F
V

W
S
X

U
J
M

T
G
B

E
D
C

T
G
B

Sheet 2

For each of the given words, you should lift and tap the corresponding finger.

Y
H
N

E
D
C

I
K

E
D
C

R
F
V

I
K

W
S
X

I
K

O
L

I
K

I
K

U
J
M

P

Q
A
Z

E
D
C

P

W
S
X

T
G
B

E
D
C

Y
H
N

O
L

E
D
C

Y
H
N

U
J
M

O
L

O
L

Q
A
Z

U
J
M

Y
H
N

U
J
M

Y
H
N

E
D
C

O
L

Y
H
N

I
K

W
S
X

Y
H
N

P

Y
H
N

Y
H
N

P

T
G
B

I
K

Q
A
Z

E
D
C

W
S
X

P

P

E
D
C

R
F
V

E
D
C

P

I
K

R
F
V

I
K

U
J
M

T
G
B

R
F
V

P

Y
H
N

W
S
X

I
K

Y
H
N

P

U
J
M

Q
A
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I
K

W
S
X

Y
H
N

O
L

E
D
C

Q
A
Z

Q
A
Z

Y
H
N

T
G
B

T
G
B

P

Y
H
N

O
L

T
G
B

Y
H
N

Sheet 2

For each of the given words, you should lift and tap the corresponding finger.

Q A Z	O L	P	E D C	W S X	E D C	W S X
Q A Z	O L	O L	O L	O L	Y H N	Q A Z
W S X	T G B	E D C	O L	I K	T G B	U J M
Y H N	W S X	T G B	I K	U J M	I K	P
U J M	R F V	I K	R F V	Y H N	E D C	P
Y H N	Y H N	I K	O L	E D C	P	Q A Z
W S X	W S X	I K	T G B	O L	W S X	O L
E D C	P	O L	O L	Q A Z	Y H N	T G B
Q A Z	R F V	E D C	W S X	O L	Y H N	Q A Z
E D C	I K	O L	Q A Z	I K	E D C	Y H N
O L	R F V	I K	P	T G B	T G B	O L
E D C	R F V	E D C	Y H N	P	R F V	Q A Z

Sheet 2

For each of the given words, you should lift and tap the corresponding finger.

Y H N	Q A Z	O L	I K	Q A Z	P	O L
W S X	Y H N	I K	P	U J M	T G B	P
P	P	W S X	E D C	W S X	U J M	T G B
O L	Q A Z	W S X	P	E D C	W S X	U J M
O L	I K	I K	U J M	Y H N	I K	P
O L	Q A Z	T G B	P	U J M	Y H N	Y H N
Q A Z	O L	O L	Q A Z	R F V	T G B	Q A Z
T G B	E D C	U J M	I K	W S X	U J M	E D C
E D C	I K	I K	I K	Q A Z	I K	O L
P	O L	I K	U J M	U J M	U J M	W S X
E D C	T G B	P	T G B	Q A Z	Q A Z	I K
E D C	P	Q A Z	R F V	R F V	P	E D C

Sheet 3

For each of the shown letters, you should pronounce the corresponding word and then lift and tap the corresponding finger (either in normal position or in shifted position, see text).

V	X	N	P	M	H	R	S	N	X	G	M	I	U
V	V	T	Y	F	Y	F	O	T	F	O	U	Y	A
B	Q	K	K	R	O	P	T	U	G	Z	V	Y	F
H	K	R	Q	T	L	Z	L	W	D	A	H	D	R
B	L	F	R	R	R	A	M	O	A	K	Z	A	X
Y	D	Z	Z	E	T	L	T	Y	L	G	T	I	O
H	V	G	S	X	X	O	N	B	O	R	A	J	M
I	J	M	K	K	K	H	E	G	J	A	U	T	I
H	R	T	V	N	D	Q	P	Q	X	Z	M	R	H
F	R	F	O	T	Z	N	D	Y	D	D	I	E	Z
Z	Z	A	N	G	B	N	R	H	X	A	G	H	U
H	Q	D	C	P	W	A	X	D	Z	X	P	J	I
F	J	L	G	F	W	E	E	C	R	D	C	Q	C
E	U	M	V	Z	X	W	O	H	E	M	P	H	P
Y	A	S	A	I	O	Z	E	Y	Z	H	E	S	H
M	C	A	J	K	Q	E	B	Y	Q	H	O	O	W

Sheet 3

For each of the shown letters, you should pronounce the corresponding word and then lift and tap the corresponding finger (either in normal position or in shifted position, see text).

V	J	P	Z	I	P	H	A	E	U	M	D	T	P
E	L	Z	X	O	B	H	H	Y	H	L	J	R	O
I	G	A	N	C	E	O	M	G	E	I	P	W	K
Y	U	I	V	L	B	H	M	Q	C	R	K	M	T
J	C	R	S	D	V	I	J	I	Y	Q	Q	F	F
N	W	L	J	B	L	S	Q	J	O	H	U	M	Z
M	B	D	D	D	R	P	O	U	N	J	L	D	Q
H	I	A	E	Y	Y	S	M	F	J	J	X	U	D
G	Y	J	Z	H	T	Y	G	R	W	Y	V	O	J
P	Y	Q	R	P	R	P	A	V	G	I	Y	C	X
B	H	K	S	K	K	D	T	U	E	T	L	X	S
D	H	U	D	A	W	N	D	C	P	R	G	K	H
W	T	B	M	M	D	X	R	M	M	R	P	A	M
G	L	O	Z	P	E	K	Q	L	A	Y	J	P	T
J	H	E	J	G	F	M	C	M	F	S	F	J	N
B	L	A	O	H	H	H	I	Q	G	H	C	V	Z

Sheet 3

For each of the shown letters, you should pronounce the corresponding word and then lift and tap the corresponding finger (either in normal position or in shifted position, see text).

N	Y	N	Y	J	Q	P	L	J	Q	J	W	S	C
L	F	F	Q	E	S	F	D	B	B	X	I	X	W
V	C	G	V	C	I	O	W	F	K	Q	F	L	T
P	V	Z	R	V	D	D	P	G	E	P	Z	D	R
W	P	T	A	X	J	X	Y	Y	N	H	N	F	M
V	Z	V	Q	E	L	X	O	Q	P	O	A	P	K
W	R	D	D	U	M	C	I	X	S	N	U	Z	K
E	X	W	I	M	G	Y	V	T	I	L	P	E	P
E	P	I	U	M	G	L	X	G	P	K	F	D	O
I	K	X	Y	L	P	N	T	H	X	V	R	S	Q
H	I	Z	J	L	O	S	D	H	Y	R	J	T	A
G	T	I	J	N	H	W	I	T	J	T	R	O	P
Q	D	L	Y	B	M	U	D	E	E	A	M	L	E
I	T	Z	L	S	R	Q	U	U	R	I	N	Y	H
S	U	T	K	M	N	V	F	I	U	T	O	I	E
N	L	L	S	K	E	Q	B	F	X	R	G	P	J

Sheet 3

For each of the shown letters, you should pronounce the corresponding word and then lift and tap the corresponding finger (either in normal position or in shifted position, see text).

P	E	S	P	D	N	W	S	Y	H	V	J	G	O
Q	N	R	Q	Q	V	N	L	E	H	T	D	L	Y
H	M	Y	R	I	F	F	F	U	C	J	A	F	A
W	E	T	T	I	Z	X	T	P	F	K	W	N	P
O	N	G	H	A	C	D	D	M	K	W	E	S	K
O	M	J	F	V	X	E	W	R	W	Q	E	T	E
L	W	I	F	G	U	C	E	A	V	F	C	L	F
S	N	P	I	C	E	J	P	B	J	A	M	S	X
H	E	V	P	X	O	T	I	G	Y	U	Z	F	H
C	N	D	X	F	G	A	R	K	D	X	J	M	D
T	F	O	L	V	F	Q	B	H	J	N	T	A	Q
J	E	P	R	H	D	U	X	I	J	P	Y	O	N
V	B	O	Q	V	V	X	C	D	J	Y	T	L	D
D	R	M	W	D	F	F	B	O	C	T	N	Y	W
M	M	L	D	U	B	C	D	A	Z	W	O	X	T
T	K	S	P	U	Z	S	L	Q	L	N	R	U	C